Computer Networks & Operation Systems

Department of Computer Education and Instructional Technology

Grup 5

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GitHub Repo Link: **https://github.com/NurullahDAGHAN/OS-Project-xMeine.git**

Network Cable Demo: Educational Computer Networks Game

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1. Project Overview

Network Cable Demo is a playable educational mobile game prototype designed to help beginner students understand basic computer networking concepts through simple interactive tasks. Instead of presenting networking topics only as abstract definitions, the project teaches them through familiar daily-life situations such as connecting a computer to a modem, choosing an IP address, and later configuring a gateway or DNS.

The project currently focuses on the first milestone of a larger educational game. The prototype introduces the first two networking concepts: physical Ethernet connectivity and local IP address selection. The long-term goal is to build a sequence of short levels that teach computer networks step by step through actions, feedback, and visual cause-and-effect.

2. Motivation

Computer networks include many concepts that can be difficult for beginners, such as Ethernet, modem, IP address, gateway, DNS, packets, Wi-Fi signal, and firewall rules. These terms are often introduced as definitions before students have an intuitive understanding of what they mean.

This project aims to make these concepts easier to understand by placing them inside small playable situations. For example, instead of only saying that an IP address identifies a host on a network, the game shows that a computer connected to a modem still needs a compatible address to communicate on the local network.

3. Educational Approach

The educational approach of the project is based on the following principles:

- Concrete examples before abstract terminology
- Action before long explanation
- One networking concept per level
- Short feedback after each user action
- Real-life analogy before formal definition
- Simple mobile-friendly interaction

This approach is intended to support students who are beginning to learn computer networks and may not yet be comfortable with technical terminology.

4. Current Prototype

The current milestone prototype includes two playable levels.

4.1 Level 1: Ethernet Connection

The first level teaches that internet access begins with physical connectivity. The player drags an Ethernet cable plug to the correct modem port. If the player drops the cable in the wrong place, the object returns to a safe state and a short hint is displayed. If the player connects the cable correctly, the modem lights turn on, the computer appears connected, and a success panel opens.

Learning outcome: The student understands that a cable creates a physical connection between the computer and the modem/router.

4.2 Level 2: IP Address

The second level teaches that a device needs a compatible IP address to communicate on a local network. The modem is represented as being on the `192.168.1.x` network. The player chooses the correct IP address from multiple options.

The current correct answer is:

```
192.168.1.24
```

Other options such as `10.0.0.9` and `172.16.4.2` are valid private IP ranges in real networking, but they do not match the current local network in this level. This allows the level to teach that an IP address must fit the network, not only look like a valid private address.

Learning outcome: The student understands that devices on the same local network should use compatible IP addresses.

5. Game Design

The game is designed as a portrait-first hyper-casual mobile experience. Each level should be short, clear, and playable within a few seconds. The game avoids heavy penalties and instead uses soft educational feedback.

Each level follows this structure:

- Title
- Instruction
- Short dialogue
- Playable task
- Hint message

- Success message
- Learning note
- Next step

The general screen layout is:

- Top: task panel and status
- Middle: interactive game scene
- Bottom: action choices or draggable objects
- Overlay: success panel

The design intentionally keeps text short so that students can focus on interaction and visual feedback.

6. Technologies Used

The prototype uses the following technologies:

Technology	Purpose
Flutter	Mobile app shell, UI panels, buttons, dialogs, and cross-platform build support
Flame	Game loop, components, scene rendering, drag interaction, and game logic
Dart	Main programming language
ValueNotifier	Lightweight UI/game state management for the prototype
SQLite	Planned local database layer for level data and player progress
PNG Assets	Planned final visual production assets

7. Technical Architecture

The current prototype uses Flutter and Flame. Flutter is responsible for the mobile application interface, while Flame is used for the interactive game scene and game components. The current level data is defined in Dart files, which is acceptable for the early prototype but becomes harder to maintain as the number of levels grows.

The planned architecture separates the app into data, game, and UI layers:

```
lib/
  data/
    database/
      app_database.dart
      database_seed.dart
      level_tables.dart
    repositories/
      level_repository.dart
      progress_repository.dart
```

```
game/
  network_game.dart
  levels/
    level_data.dart
ui/
  game_hud.dart
  ip_task_panel.dart
```

This structure keeps the database layer separate from the Flame game scene. The game scene should receive already loaded level data from a repository or controller instead of querying SQLite directly.

8. Database Design

SQLite is planned as the local database for the project. The purpose of SQLite is to store level definitions, level order, educational messages, answer options, unlock status, completion state, and attempt counts.

SQLite should store structured data such as:

- Level list
- Level order
- Level title and instruction
- Educational messages
- Task type
- Level objects and positions
- Answer options
- Player progress
- Completion state
- Attempt count

SQLite should not store large image or audio files. Visual and sound assets should remain in the app bundle, while SQLite stores references or keys.

The proposed database includes tables such as:

- `levels`
- `level_objects`
- `cable_goals`
- `ip_selection_goals`
- `ip_selection_options`
- `player_progress`

The production unlock logic is planned as follows: only the first level is unlocked at the beginning, and completing one level unlocks the next level.

9. Current Milestone Status

At the current milestone, the project includes:

- Flutter + Flame prototype

- Portrait-friendly layout direction
- Level 1: Ethernet cable connection
- Level 2: IP address selection
- HUD and success panel design
- Basic visual feedback
- Documentation for product vision, game design, technical plan, SQLite model, and roadmap

10. Future Work

The next development steps are:

1. Improve the visual consistency of the first two levels.
2. Add SQLite implementation for local level and progress storage.
3. Add a level selection screen with locked/unlocked states.
4. Add Level 3 about the default gateway.
5. Add Level 4 about DNS.
6. Replace code-drawn placeholders with final PNG assets.
7. Add sound effects and haptic feedback.
8. Track student progress such as attempts, hint usage, and completion state.

11. Risks and Limitations

The main risks of the project are:

- Adding too much text and making the game feel like a lecture
- Making networking concepts too abstract too early
- Having inconsistent visual style
- Adding database complexity before level design is stable
- Depending too much on final assets before the gameplay loop is complete

To reduce these risks, the project keeps each level small, focuses on one concept at a time, uses short instructions, and prioritizes phone-screen readability.

12. Conclusion

Network Cable Demo is an educational game prototype that teaches beginner-level computer networking concepts through simple mobile interactions. The current milestone demonstrates the core learning loop with Ethernet connection and IP address selection levels. The project uses Flutter and Flame for the playable prototype and plans to use SQLite for structured local level and progress data.

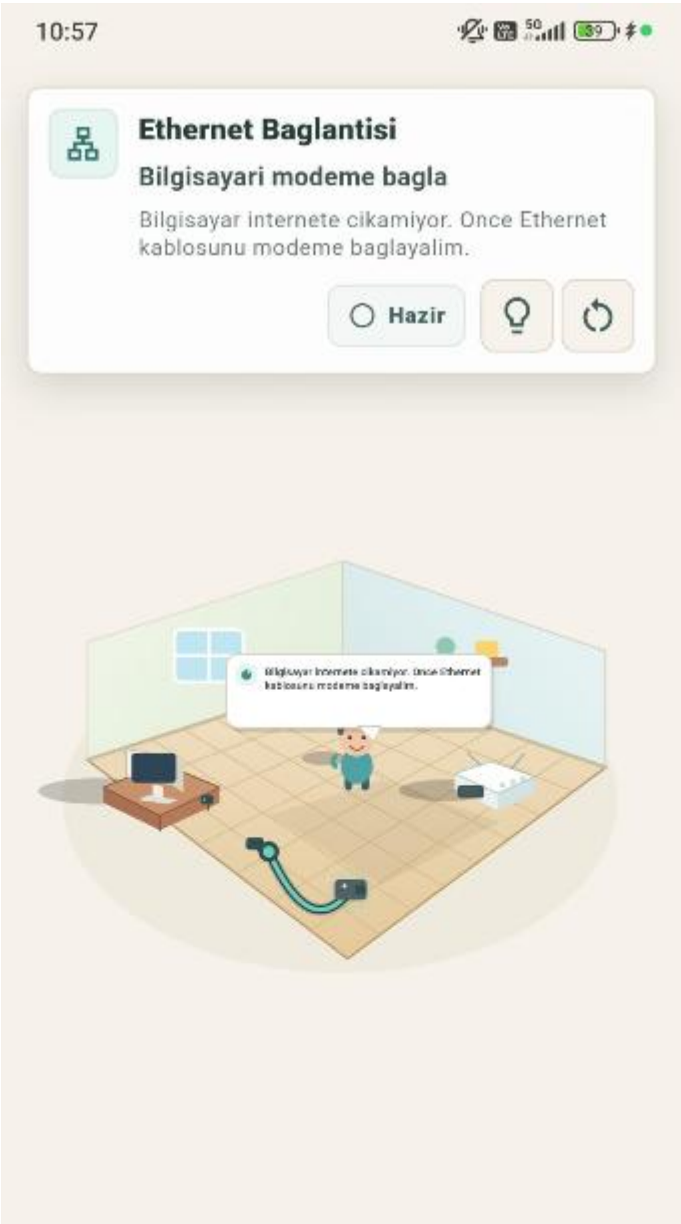
The long-term goal is to expand the prototype into a sequence of short networking puzzles that help students build intuition before learning formal computer network theory.

Appendix A: Repository Contents

The repository includes:

- Source code for the Flutter/Flame prototype
- Documentation files
- Product vision document
- Game design document
- Technical plan
- SQLite data model plan
- Roadmap
- README file

Appendix B: Screenshots



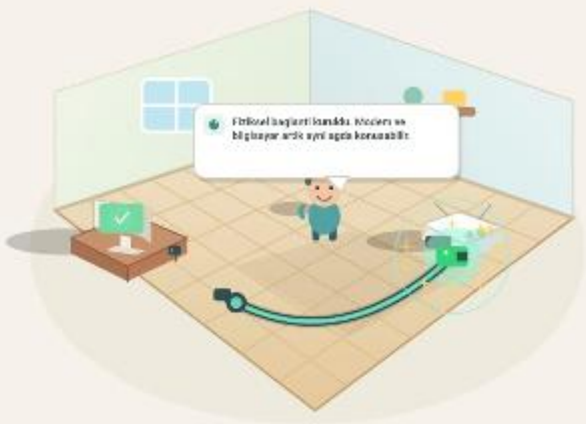
10:57

Ethernet Baglantisi

Bilgisayari modeme bagla

Fiziksel baglanti kuruldu. Modem ve bilgisayar artik ayni agda konusabilir.

Baglandi



11:00

Ethernet Baglantisi

Bilgisayari modeme bagla

Fiziksel baglanti kuruldu. Modem ve bilgisayar artik ayni agda konusabilir.

Tamamlandi

Basarili!

Basarili! Bilgisayar artik modeme bagli. Bir sonraki adimda IP adresini ogrenecegiz.

Ogrenilen konu

Ethernet kablosu, bilgisayar ile modem arasinda fiziksel ag baglantisi kurar.

Sonraki adim

Bir sonraki bolumde bilgisayarin ag uzerinde IP adresi almasini ele alacagiz.

Tekrar dene